

PAPER_518 — DPM-Unified Inertia, Centripetal, and Centrifugal Forces: Resolving the Classical Conundrum

Unified Quantum Field Framework (UQFF)

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§1 — Abstract

Classical mechanics leaves an unresolved conundrum: inertia is treated as an intrinsic resistance property (Newton's First Law), centripetal force as a real inward constraint force, and centrifugal force as a fictitious pseudo-force appearing only in non-inertial frames. None provides a causative origin for mass. In the Star-Magic UQFF framework all three are **pure forces**, emergent from DPM reaction in 26D layered shells. Their equilibrium defines mass occurrence without invoking any intrinsic property.

§2 — Classical Conundrum

Force	Classical Status	Problem
Inertia (F_{inert})	Intrinsic resistance, not a force	Has no dynamical origin; Mach's Principle unresolved
Centripetal (F_{centrip})	Real inward force (tension, gravity)	Agent-dependent; not fundamental

Force	Classical Status	Problem
Centrifugal (F_{centrif})	Fictitious pseudo-force	Vanishes in inertial frame; not a pure force

Result: Mass cannot be derived from these three; all remain postulates.

§3 — DPM-Unified Definitions

3.1 — Inertial Force

$$\boxed{F_{\text{inert}}} = -\frac{\partial \text{bigl(DPM}_{\text{react}} \cdot \text{ShellEnergy}\bigr)}{\partial v^{26}} \cdot t_{\text{neg}}$$

- $\text{DPM}_{\text{react}}$ is the DPM reaction strength (see PAPER_516 §3.2)
- ShellEnergy is the layer-integrated radiance
- $t_{\text{neg}} < 0$: the negative time factor ensures $F_{\text{inert}} > 0$

(resists velocity change)

- **Mass occurrence:** $M = F_{\text{inert}} / a^{26}$

3.2 — Centripetal Force

$$\boxed{F_{\text{centrip}}} = \frac{\text{DPM}_n(\text{SCm}) \cdot \omega_{\text{CW}}^2 \cdot r^{\text{layer}}}{1 + \Delta_{\text{dil}}}$$

- DPM north (SCm) component pulls radiance inward (CW direction)
- Dilation correction $(1 + \Delta_{\text{dil}})^{-1}$: relativistic shells

have weaker centripetal grip (consistent with frame dragging)

- Maintains layered shell coherence; in proto-H shells, condenses smalls into nuclei

3.3 — Centrifugal Force

$$\boxed{F_{\text{centrif}}} = \text{DPM}_s(\text{UA}') \cdot \omega_{\text{CCW}}^2 \cdot r^{\text{layer}} \cdot t_{\text{neg}}$$

- DPM south (UA') component pushes radiance outward (CCW direction)
- $t_{\text{neg}} < 0$: the negative factor makes F_{centrif}

oppose centripetal (correct sign convention)

- Drives expansion toward Big Bang speed via buoyant radiance outflows

§4 — Mass Equilibrium Condition

$$\boxed{F_{\text{inert}} = F_{\text{centrip}} - F_{\text{centrif}}}$$

Mass occurs at the point where inertial resistance equals the net shell compression force. The 26D acceleration is:

$$a^{26} = \frac{F_{\text{centrip}} - F_{\text{centrif}}}{M} = \frac{F_{\text{inert}}}{M}$$

Dual existence symmetry: $F_{\text{centrif,one}} = -F_{\text{centrif,opp}}$

One-side centrifugal drives expansion; opposite side infers inward compression — the universe self-balances across the 26D egg.

§5 — Resolution of the Classical Conundrum

Classical Problem	Star-Magic Resolution
Inertia has no origin	F_{inert} emerges from DPM grinding via $\partial/\partial v^{26}$
Centrifugal is fictitious	F_{centrif} is a real DPM south force (UA' CCW push)
Mass origin unknown	$M = F_{\text{inert}} / a^{26}$ — mass is equilibrium, not intrinsic
Mach's Principle	Inertia encoded in global DPM layered shell reactions

The conundrum dissolves as a downward projection artefact: 26D grinding unifies all three, and in the 3D observable slice they appear distinct only because the layered structure is not directly visible.

§6 — Atomic Mass Uniqueness

Non-repeating quantum fingerprints per atom arise because each atom's F_{inert} is set by its specific layer index l , its local (DPM_n, DPM_s) reaction state, and the t_{neg} dilation at that spacetime location. No two atoms share the same radiance history → unique mass values per atom confirmed by the Standard Model mass spectrum.

§7 — Canonical Constants

Symbol	Value	Units
\$DPM_n\$ (normalised)	1.0	—
\$DPM_s\$ (normalised)	0.85	—
ω_{CW}	1.2×10^{10}	rad/s
ω_{CCW}	8.3×10^9	rad/s
κ	5×10^{-4}	—

§8 — Conclusion

Inertia, centripetal, and centrifugal forces are unified as pure emergent forces from DPM-layered radiance in 26D shells. Their equilibrium condition $F_{\text{inert}} = F_{\text{centrip}} - F_{\text{centrif}}$ defines mass occurrence without invoking intrinsic properties. The 3,000-year classical conundrum of the status of centrifugal force is resolved.

See also: PAPER_516 (DPM Shell Radiance), PAPER_517 (Negative Time Proof), PAPER_519 (Shell Radiance Prototype), PAPER_520 (Session 140 Hub).